

BirdsEye: Enabling Practical Geospatial-AI via RTK Geotagging Annotation Strategies



Morgan Masters

mwmaster@ucsc.edu

<https://morganmasters.github.io/>

(Image) OpenDroneMap texturized pointcloud of UCSC Farm Haybarn Field.



UNIVERSITY OF CALIFORNIA
SANTA CRUZ

UAV-based AI vision models can automate land survey video analysis.

- **Uncrewed Aerial Vehicles** (UAVs) made aerial imagery cheap.
- Large-area monitoring? ***Too much video to watch.***
 - ✓ **Artificial Intelligence** (AI) automates nontechnical tasks.
 - ✗ **New AI questions** from every user.

Problem: New AI training requires lots of annotated data.

- Online datasets? *Maybe.*
- Manual annotation is *slow.*
- Modern annotation services:
 - *Outsource* the problem.
 - Depend on *foundation models.*
 - *Cloud services.*

Spend lots of labor?

Spend extra money?

Think of small or rural users...

*What if you could train custom AI for UAVs from **GPS-tagged field observations**?*

*Can I use GPS
to break the image annotation bottleneck?*

- Place GPS waypoints on training targets.
- UAV automatically surveys GPS waypoints.
- Associate GPS waypoints with image pixels.

I. Embedded System Design

II. Operational Concepts, Software Design, and Performance Limitations

III. Results

Burrow Detection Case Study

We designed ***BirdsEye*** on these principals:

Open Source

Transparent, customizable, free.

Data Privacy

Data and model are yours and only yours.

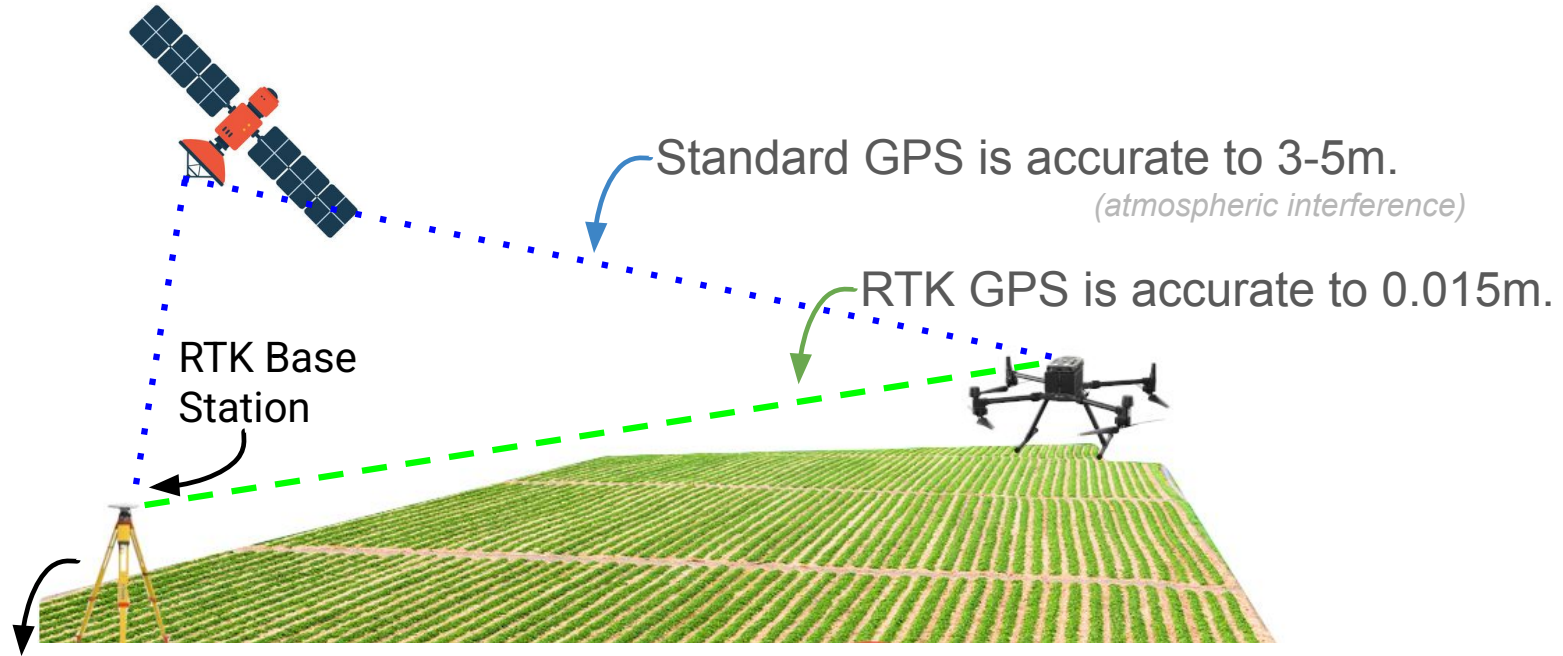
Right to Repair

Trustworthy, reparable, modular.

I. Embedded System Design

- RTK Basics
- System Overview

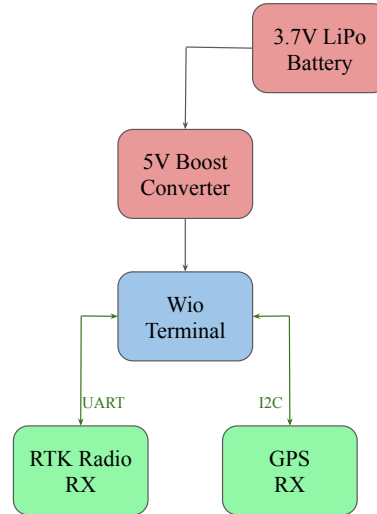
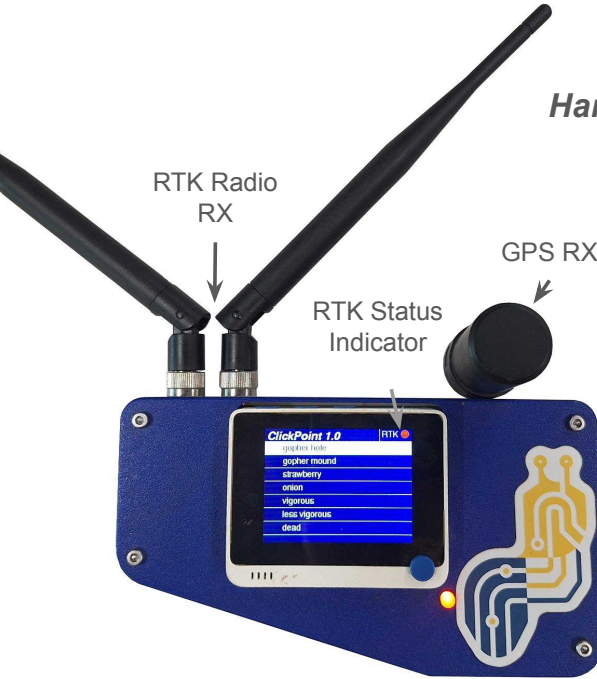
Real Time Kinematic Positioning (RTK) Basics



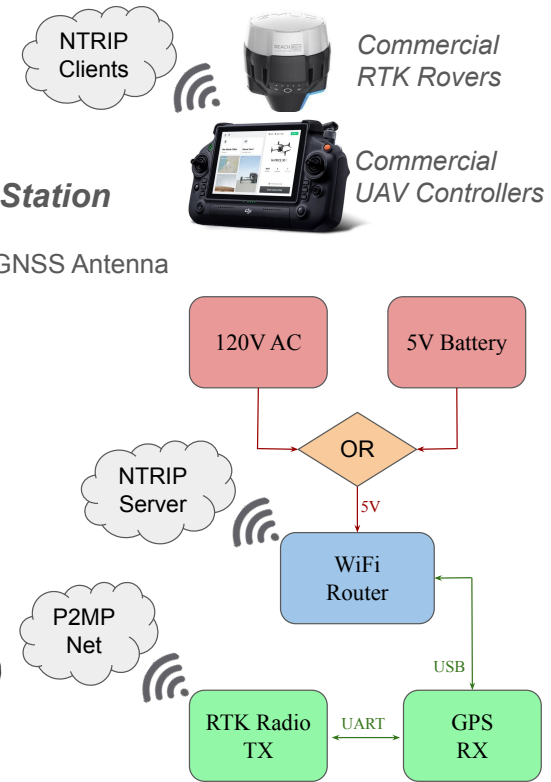
1. Establish a GPS fix, then refine position estimates by averaging over long times.
2. Use estimate as reference to broadcast high-precision corrections to listeners.

Custom RTK Annotation System Design

Handheld RTK Annotator



Portable Base Station



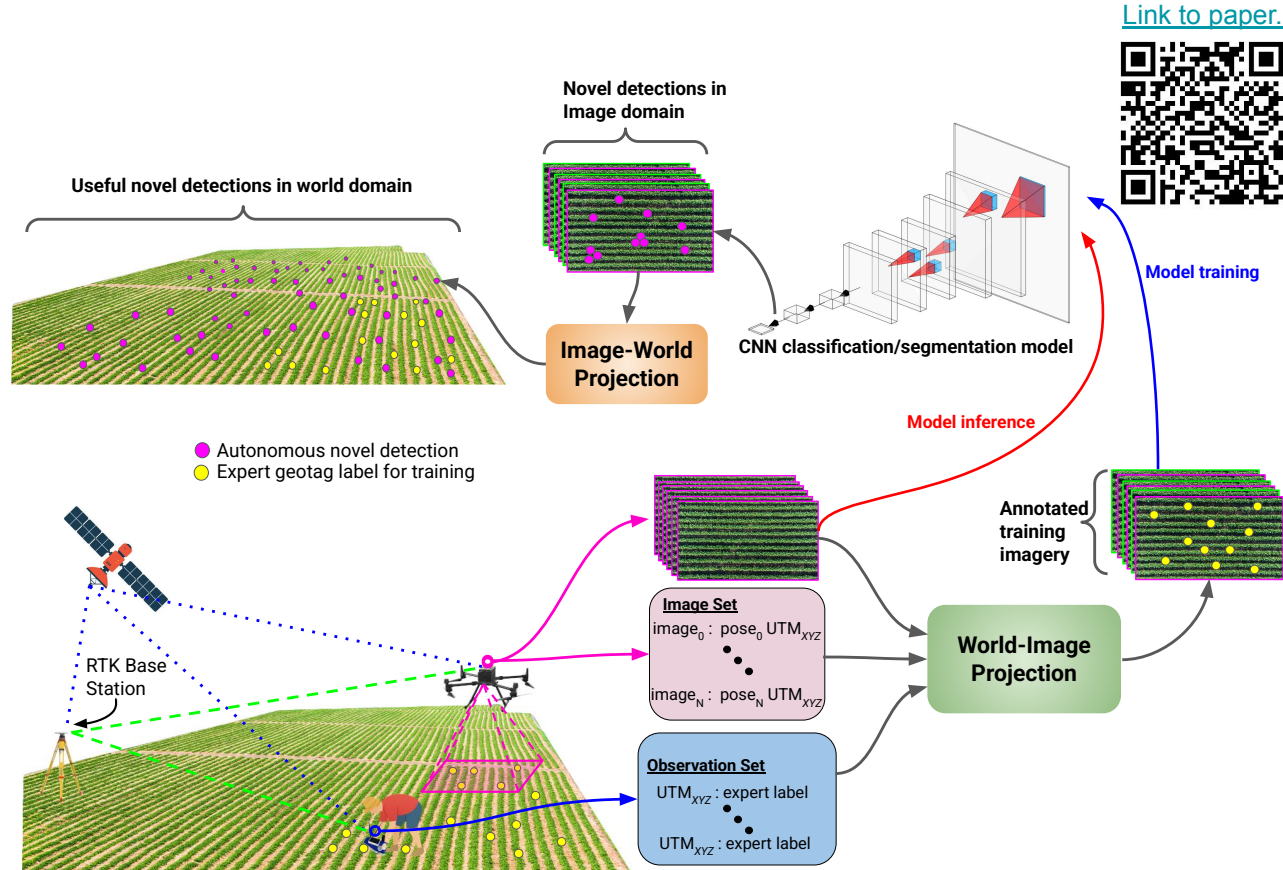
> \$1500USD cheaper than commercial products.
Off-the-shelf components, same performance.

System Overview

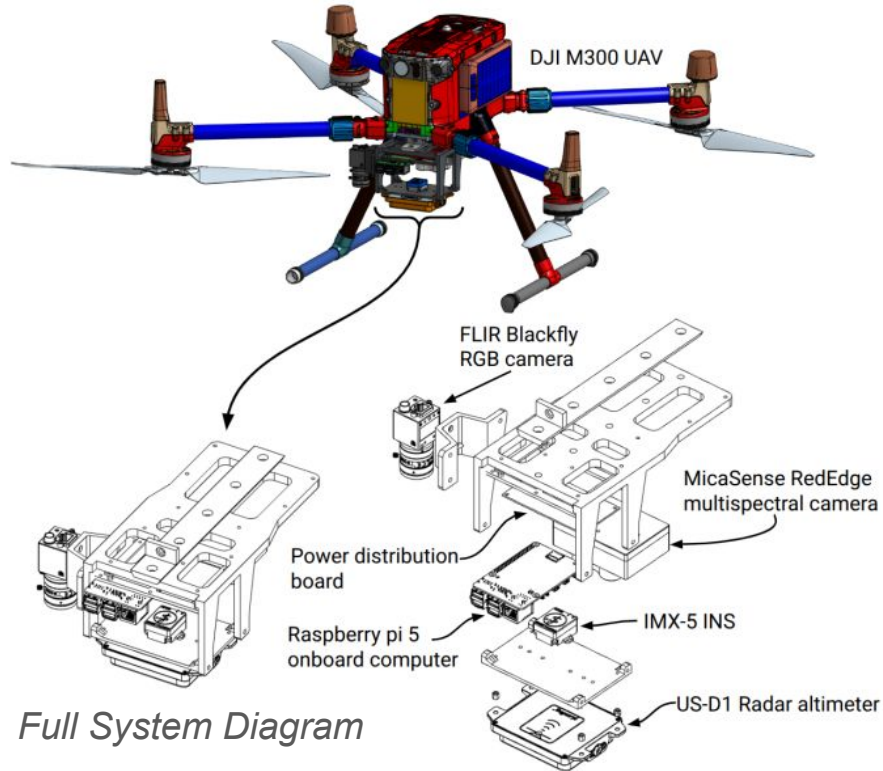
“Enabling Practical
Geospatial-AI via **RTK**
Geotagging Annotation
Strategies”



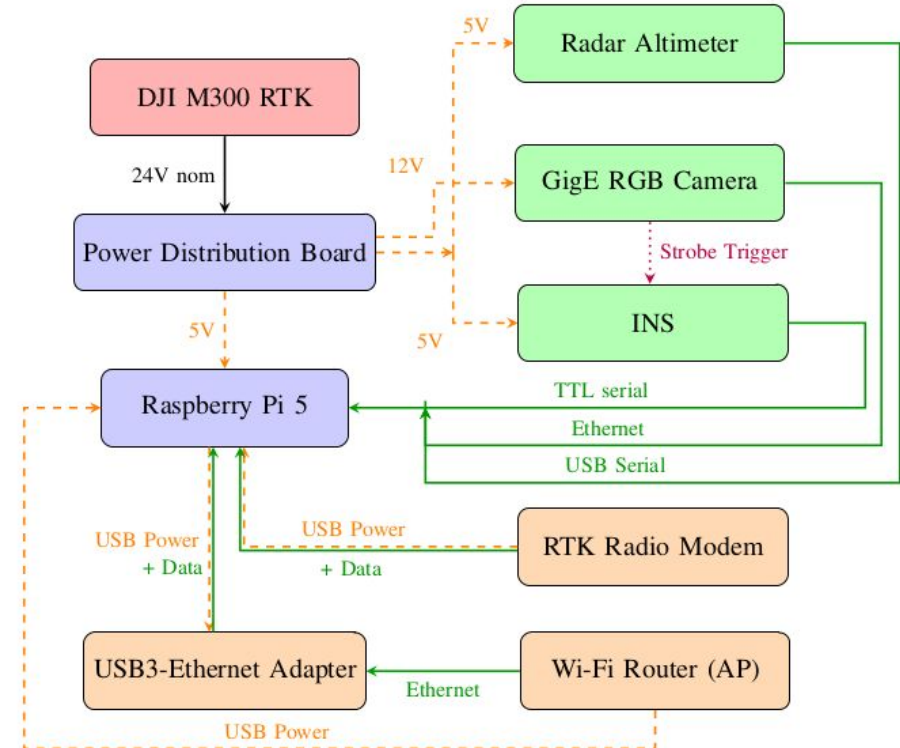
A partnering farmer learning to use BirdsEye on fresh gopher holes.



Camera System Design



System Block Diagram, as flown in our case study.



II. Operational Concepts, Software Design, and Performance Limitations

- Geotag Detection
 - Projective Geometry, Rigid-body Transforms
- Soft Labels
 - Unavoidable Projection Error

Geotag Detection

Transforming between the world and imagery.

Projective Geometry

- Pinhole Camera Model:
 - Linear, matrix K
- Real lenses distort the projection
 - Undo via Brown-Conrady (radtan) model
- Project to and from the standard world reference via rigid-body transforms, T:

$\mathbf{p}$ pixel coordinates

$\mathbf{K}$ camera matrix

$\mathbf{T}_{CI}$ transform from INS to camera

$\mathbf{T}_{IW}$ transform from world reference to INS

$\mathbf{T}_{CW}$ transform from world reference to camera

$\mathbf{x}$ world reference coordinate

$$\mathbf{p} = \mathbf{K} \mathbf{T}_{CI} \mathbf{T}_{IW} \mathbf{x}$$

The ideal case (perfect lens)...

$$\mathbf{p} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_1 & r_2 & r_3 & t_x \\ r_4 & r_5 & r_6 & t_y \\ r_7 & r_8 & r_9 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

$\mathbf{K}$ (blue arrow) $\mathbf{T}_{CW}$ (red arrow)

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x x_c / z_c + c_x \\ f_y y_c / z_c + c_y \\ 1 \end{bmatrix} \quad (\text{if } z \neq 0)$$

(projection)

...Real case

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x_c / z_c \\ y_c / z_c \end{bmatrix} \quad (\text{if } z \neq 0)$$

(normalization)

$$\begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} x'(1 + k_1 r^2 + k_2 r^4) + 2p_1 x' y' + p_2 (r^2 + 2x'^2) \\ y'(1 + k_1 r^2 + k_2 r^4) + p_1 (r^2 + 2y'^2) + 2p_2 x' y' \end{bmatrix}$$

(distortion correction)

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x x'' + c_x \\ f_y y'' + c_y \\ 1 \end{bmatrix} \quad (\text{if } z \neq 0)$$

(projection)

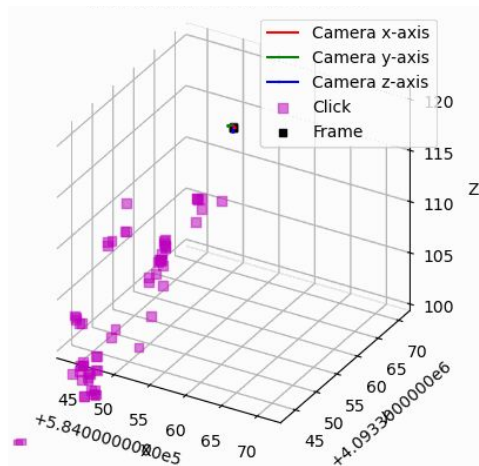
Geotag Detection

Detection is a set of cross-products:

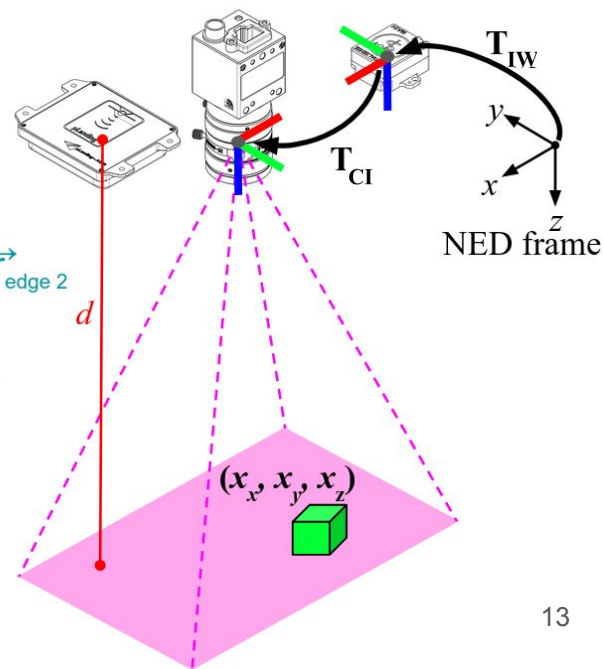
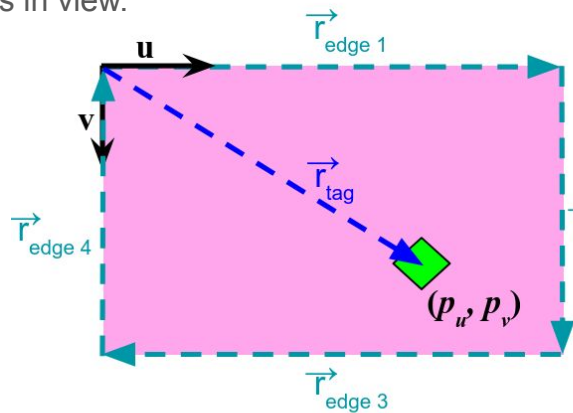
- If a product is positive, then the geotag is “on the right side”.
- If negative, then not.
- If all 4 products are positive, then the tag is in view.

$$\mathbf{p} = [\mathbf{u} \ \mathbf{v} \ 1]^\top = \mathbf{K} \mathbf{T}_{\text{CI}} \mathbf{T}_{\text{IW}} \mathbf{x}$$

$$\mathbf{r}_{\text{edge}} \times \mathbf{r}_{\text{tag}} = u_{\text{edge}} v_{\text{tag}} - u_{\text{tag}} v_{\text{edge}}$$



A GIF of the process running on real flight data.
Geotags ("Clicks") are represented as **pink squares**.
Frame corners are represented as **black squares**.

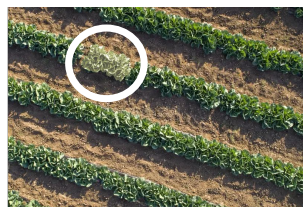


Soft Labels

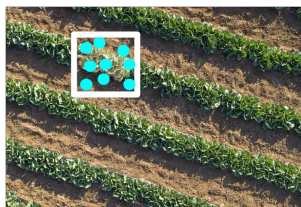
Adapting to projection performance limitations.

Testing BirdsEye

Bad calibration increases projection errors, leading to **false annotations**.



Target patch.

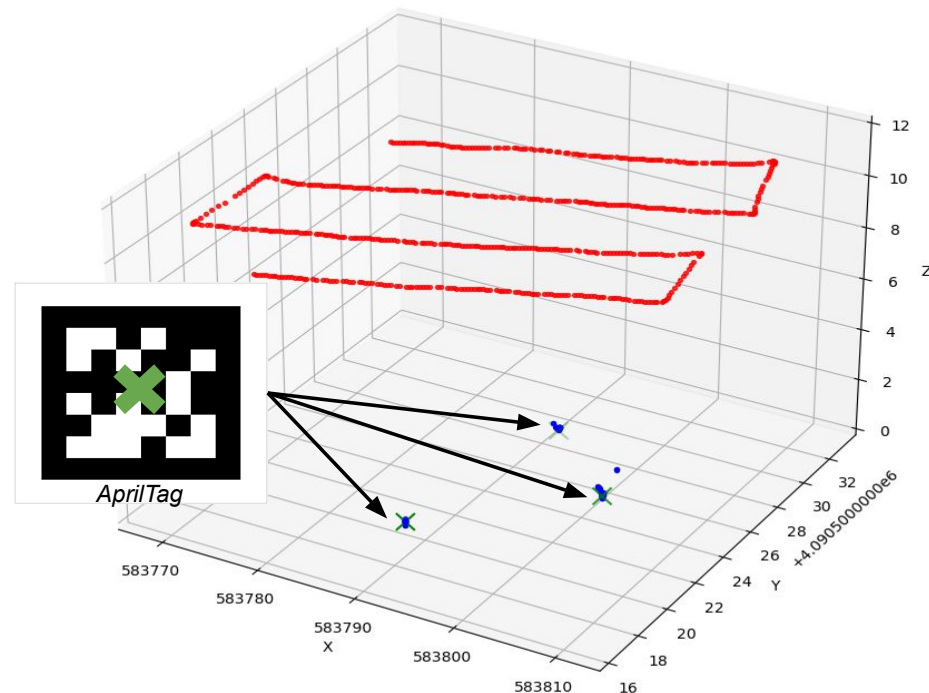


Better calibration.



Worse calibration makes false annotations.

Red markers show camera position.
Blue markers show BirdsEye projections.
Green 'X' markers show the true locations of each RTK point.

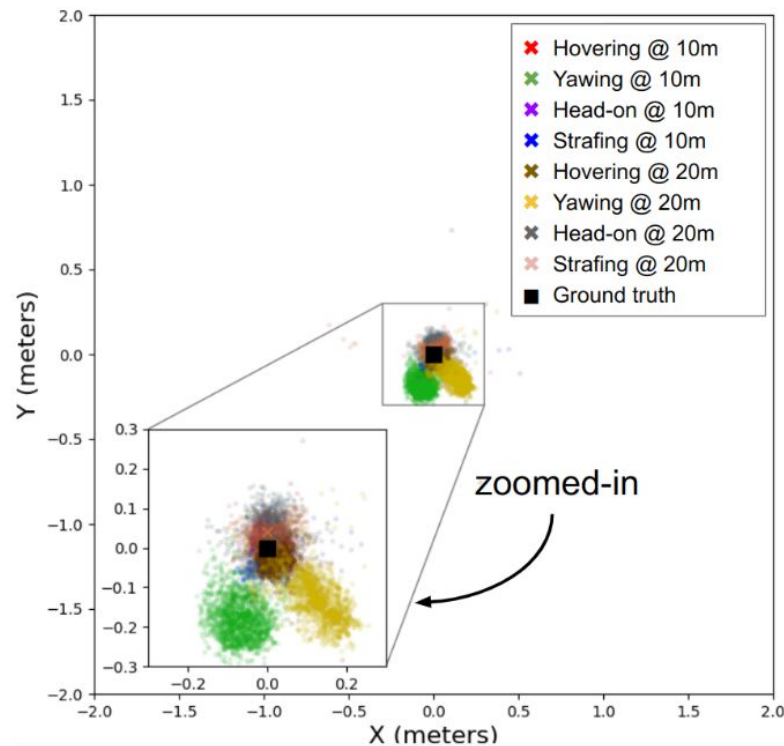


Reprojection Error

4 flight patterns, 2 altitudes.

Takeaways:

- Avoid sustained yawing.
- 6cm (20 pixel) mean error.
- 4cm (12 pixel) standard deviation.

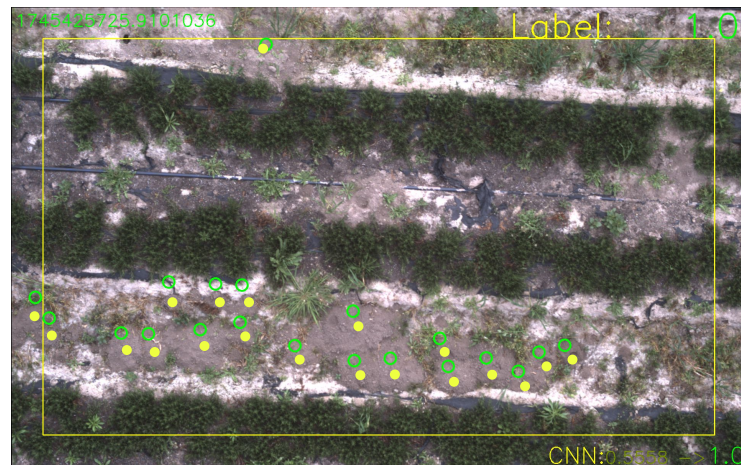


Soft Labels

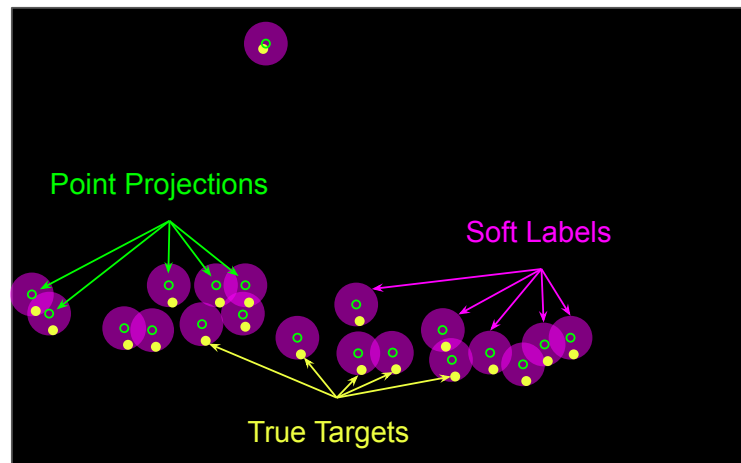
Applied labels are Gaussian patches.

- Accounts for projection errors:
 - Given a projection error model, convert high-confidence point labels to spatially-distributed patches.
 - If projection errors cause a point label to misalign with the target feature, the corresponding soft label can still cover the target.
- **Consequence:** Unconfident labels lead to unconfident learning.
 - In practice, retains sensitivity to the target textures in the hard task of burrowing pest mound detection.

We set the soft label radius to 3σ , with $\sigma = 10\text{px} \Rightarrow 60\text{px}$ diameter.



Training Input



Training Label

III. Results

A Case Study on Burrowing Pest Detection.

Deploying BirdsEye

Image: Google Earth

Flight Environment: Haybarn Field

III. Results



Step 1: Annotate

Image: Google Earth

Flight Environment: Haybarn Field

III. Results



Step 2: Optimized flight

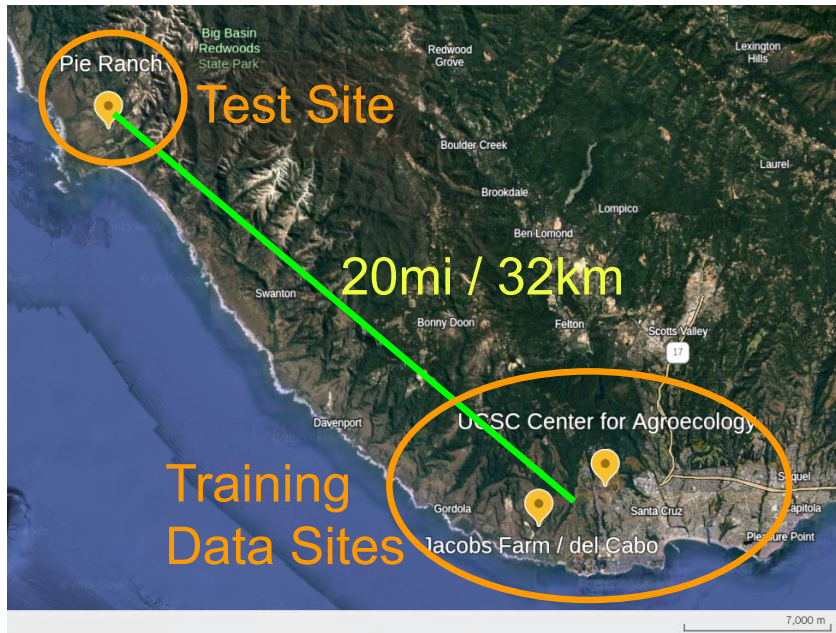
Image: Google Earth

Flight Environment: Haybarn Field

III. Results



“Enabling Practical Geospatial-AI via RTK Geotagging Annotation Strategies”



	Workers	Frames	Labor Hours	Rate (s/frame)↓
Manual	15	1,500	4:48:52	11.55
BirdsEye	2	12,524	11:22:52	3.27

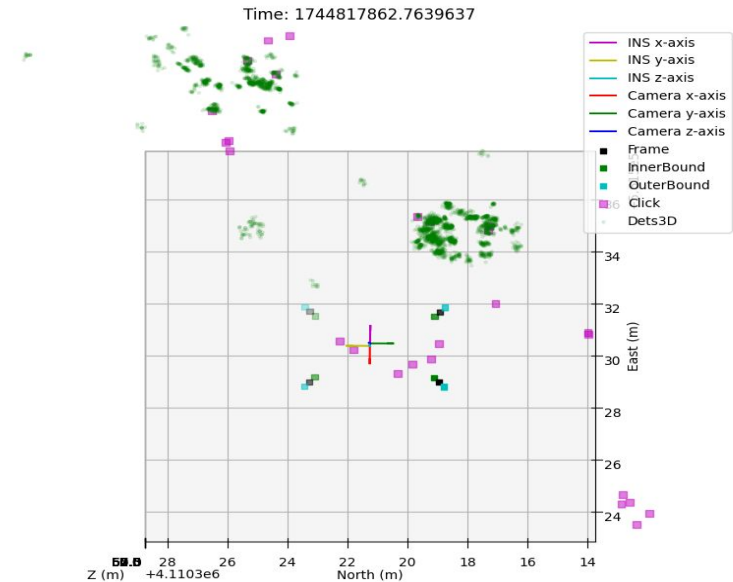
$$11.55 / 3.27 = 3.53$$

3.5x annotation rate

$$3.53 \times 7.50 = 26.49$$

26x worker output

“Enabling Practical Geospatial-AI via RTK Geotagging Annotation Strategies”



	Count	Ratio
Geotags Placed (<i>Ground Truth</i>)	161	—
Geotagged Burrows Viewed	141 of 161	87.6%
Geotagged Burrows Detected	118 of 141	83.7% (<i>Recall</i>)

Sustainably monitoring low-power, in-situ sensor arrays via field robotics.

SCAN ME



Josephson lab website: sensors.soe.ucsc.edu

McGuire lab website: hare.engineering.ucsc.edu



Morgan Masters, PhD Candidate, University of California, Santa Cruz
mwmaster@ucsc.edu, <https://morganmasters.github.io/>



Colleen Josephson, Asst. Prof, Scalable Sensing Systems for Sustainability.
cjosephson@ucsc.edu, [http\(s\)://cjosephson.net](http(s)://cjosephson.net)



Steve McGuire, Asst. Prof, Human Aware Robotics Exploration (HARE) Lab
steve.mcguire@ucsc.edu

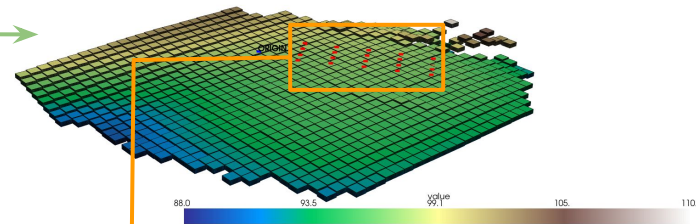
What's Next? Digital Twin Capture for Informative Path Planning over WSNs

UAV Imagery Capture



Voxelize

“Digital Twin”

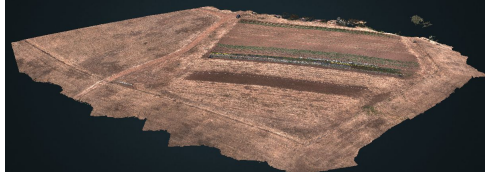


Plan UAV trips over the WSN to collect the best subset of data for soil property mapping.



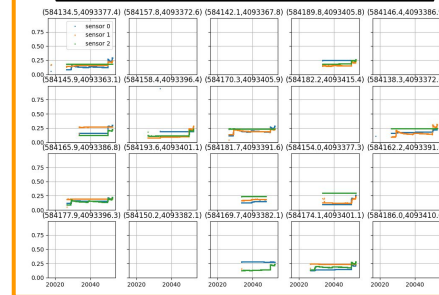
OpenDroneMap™

Georeferenced 3D
Reconstruction



Pointcloud or Orthophoto

Live Georeferenced Sensor Feeds



Sustainably monitoring low-power, in-situ sensor arrays via field robotics.

SCAN ME

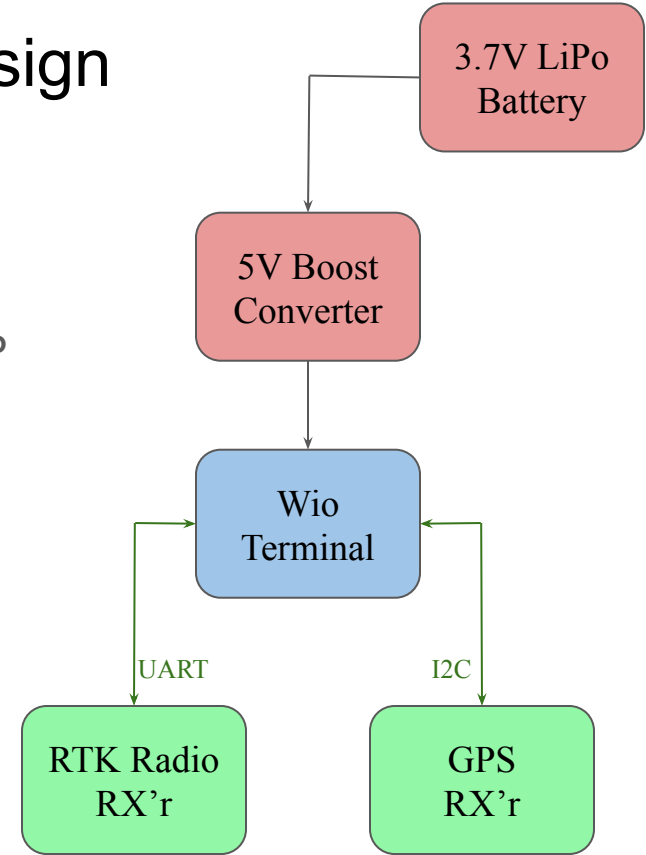
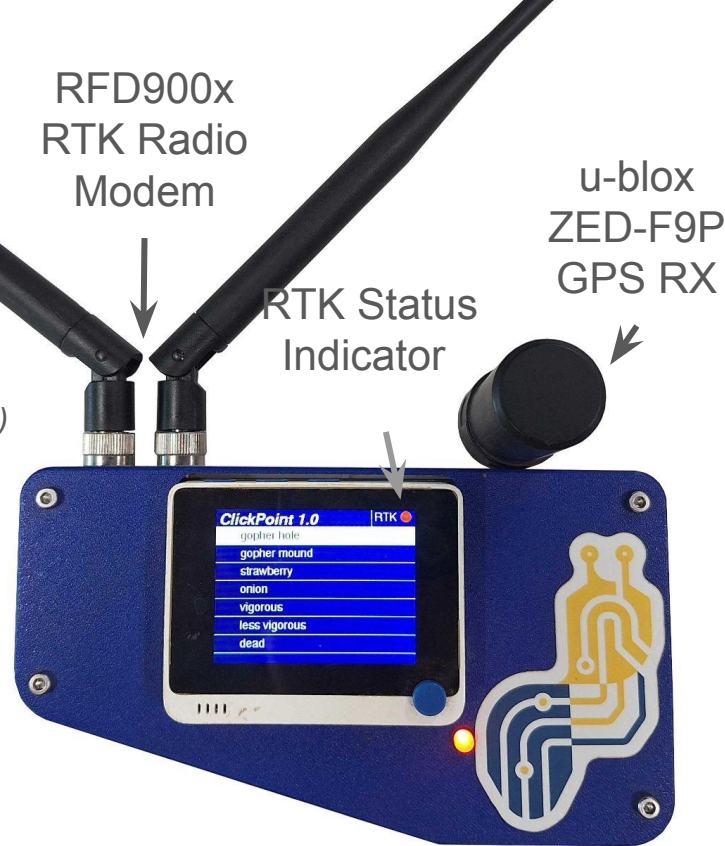


- 🌱 Josephson lab website: sensors.soe.ucsc.edu
McGuire lab website: hare.engineering.ucsc.edu
- 🌱 *Morgan Masters*, PhD Candidate, University of California, Santa Cruz
mwmaster@ucsc.edu, <https://morganmasters.github.io/>
- 🌱 *Colleen Josephson*, Asst. Prof, Scalable Sensing Systems for Sustainability.
cjosephson@ucsc.edu, [http\(s\)://cjosephson.net](http(s)://cjosephson.net)
- 🌱 *Steve McGuire*, Asst. Prof, Human Aware Robotics Exploration (HARE) Lab
steve.mcguire@ucsc.edu

Custom Handheld RTK Annotator Design

Features:

- Custom PCB
- Custom GUI
- u-blox API
- SD card for data portability
 - Rename labels (*labels.txt*)
 - Data record (*data.csv*)



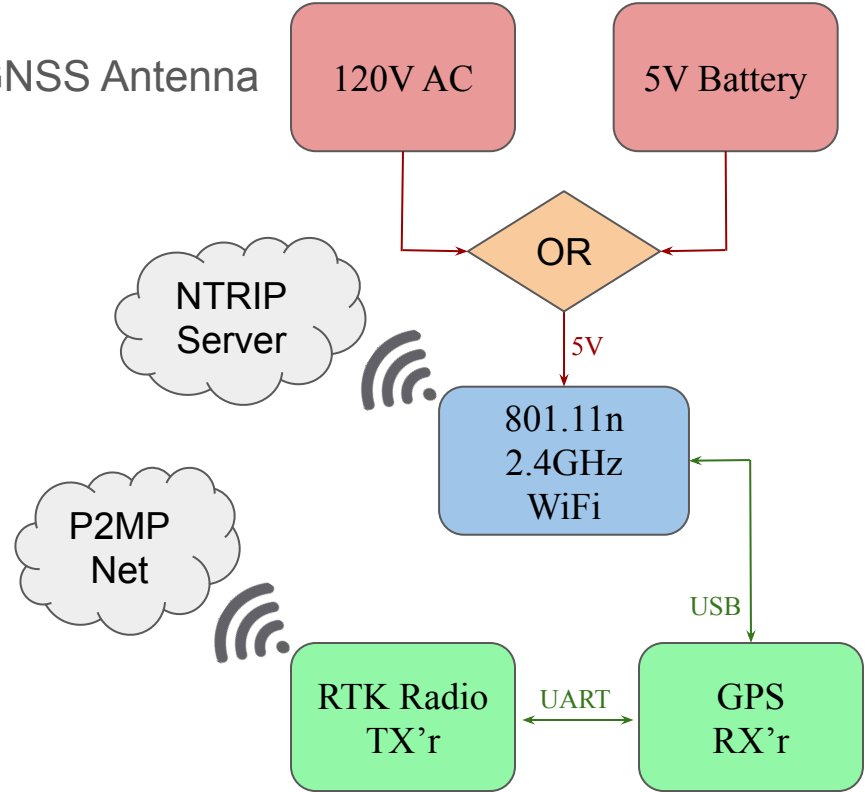
RTK Base Station Design

Features:

- Flexible Power
 - AC 120V - or -
 - Internal 5V Battery
- Flexible RTK Service
 - Point-to-Multipoint (P2MP) Radio Network
 - WiFi-hosted NTRIP Server



Gen. 1: Portable Base Station

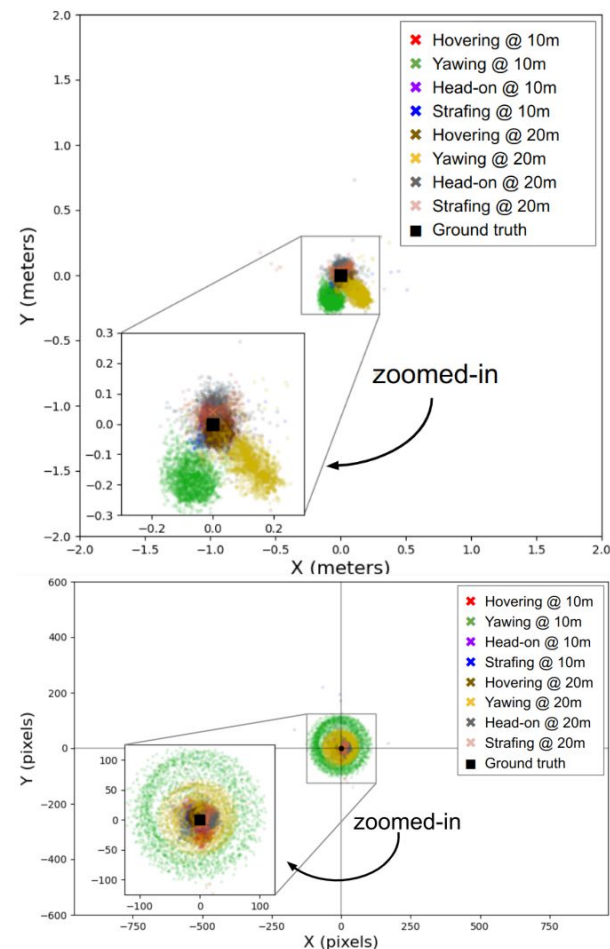


Reprojection Error

NORMS AND UNCERTAINTIES OF ERROR VECTORS FROM
GEOREFERENCING BENCHMARK TESTS AT DIFFERENT FLIGHT
MANEUVERS AND ALTITUDES, REPORTED IN METERS AND PIXELS.[†]

Altitude	Flight Maneuver	Error Norm		Standard Deviation	
		Meters	Pixels	Meters	Pixels
10m	Hovering	0.038	18.8	0.019	9.7
	Yawing	0.191	82.2	0.052	22.1
	Head-on	0.045	19.6	0.024	10.4
	Strafing	0.062	27.4	0.044	19.7
20m	Hovering	0.043	9.8	0.037	8.4
	Yawing	0.179	41.1	0.065	14.8
	Head-on	0.081	18.6	0.033	7.6
	Strafing	0.074	16.9	0.069	15.7

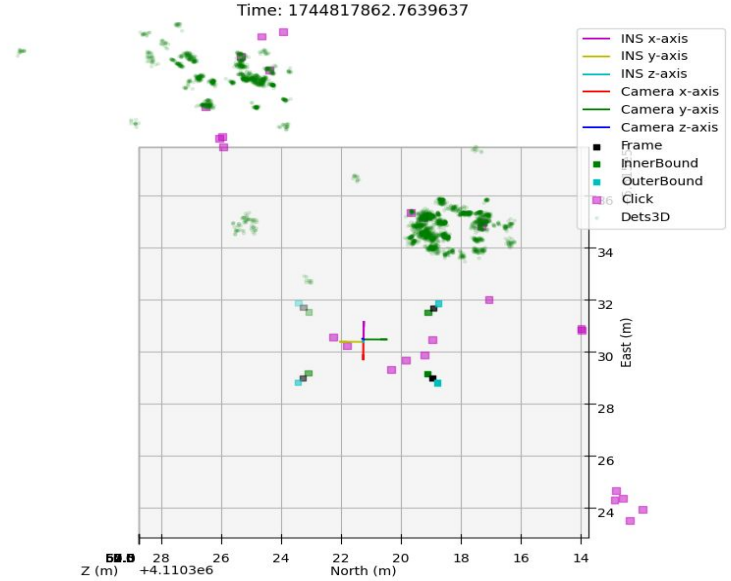
[†]The data of this table gives estimates of our effective GSD at both test altitudes: 0.002219 ± 0.000138 m (2.219 ± 0.138 mm) @ 10m; 0.004376 ± 0.000021 m (4.376 ± 0.0212 mm) @ 20m.





SUMMARY OF THE GROUND TRUTH SURVEY AT PIE RANCH, IN-FLIGHT GEOTAG INTERACTIONS, AND DETECTION RESULTS.

	Count	Ratio
Geotags Placed (<i>Ground Truth</i>)	161	—
Geotagged Burrows Viewed	141 of 161	87.6%
Geotagged Burrows Detected	118 of 141	83.7% (<i>Recall</i>)
Detection Clusters	318	—
Clusters Matched to Geotags	88 of 251	35.1% (<i>Precision</i>)



RTK Reproducibility

Can I reuse my geotags between outings?

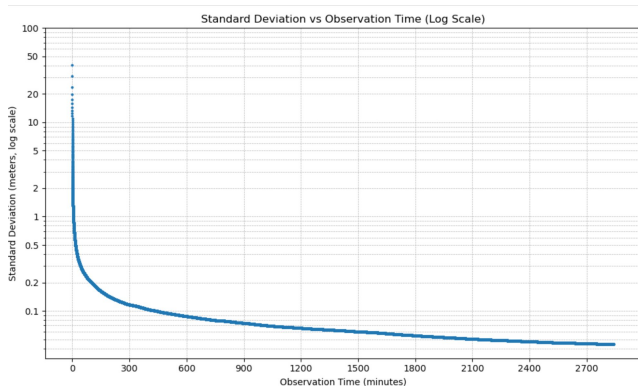
RTK Reproducibility

Tests on the RTK solution for survey-in reproducibility revealed that our current survey-in parameters prioritize **speed** over **accuracy**.

The experimental structure is 4 treatments of the RTK base station/clicker system:

1. Power cycle nothing.
2. Power cycle the base station only.
3. Power cycle the clickers only.
4. Power cycle both.

In each treatment, we collect 30 clicks at a reproducible position (*alignments with spray-painted marks on the ground for base station and clicker*) before iterating on the treatment.



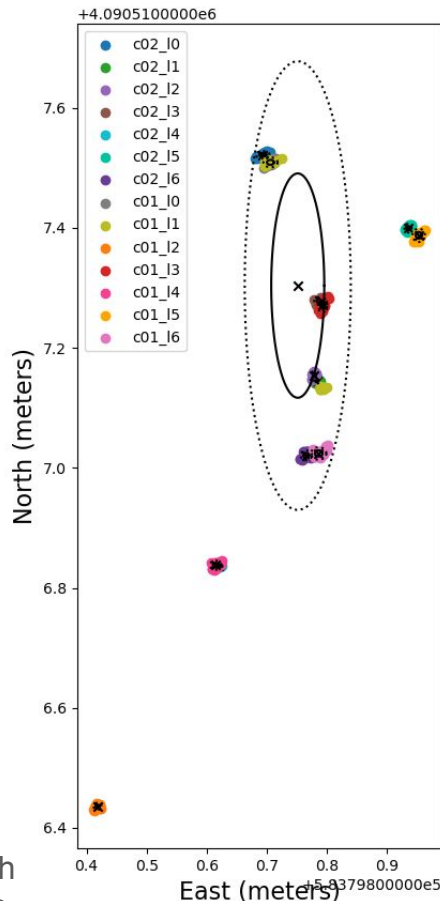
(LEFT) The survey-in accuracy-versus-time tradeoff.

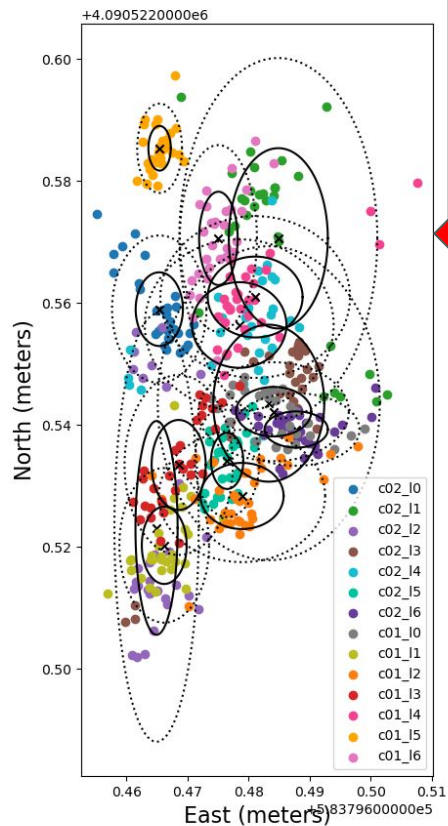
$$\sigma = 3.000\text{m} \Rightarrow t = 60\text{s}$$

$$\sigma = 0.200\text{m} \Rightarrow t = 100\text{s}$$

$$\sigma = 0.043\text{m} \Rightarrow t = 2848\text{s}$$

(RIGHT) Treatment 4: the experiment which revealed our problem.



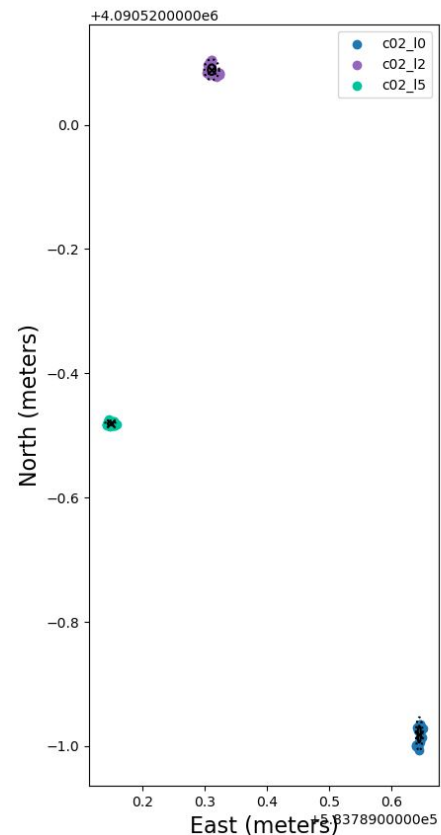
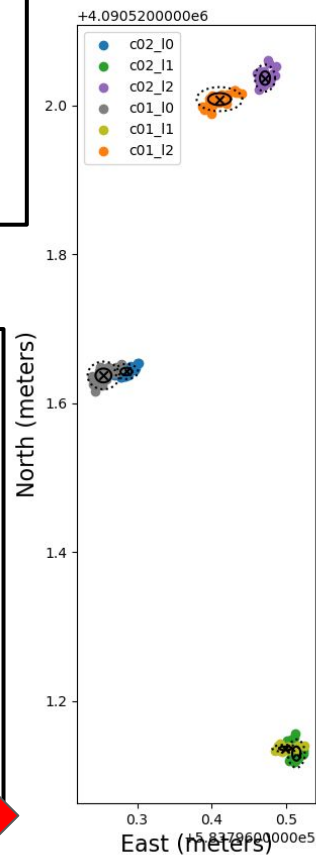


Results of treatments 1 (3 sets of 30) and 3 (4 sets of 30).

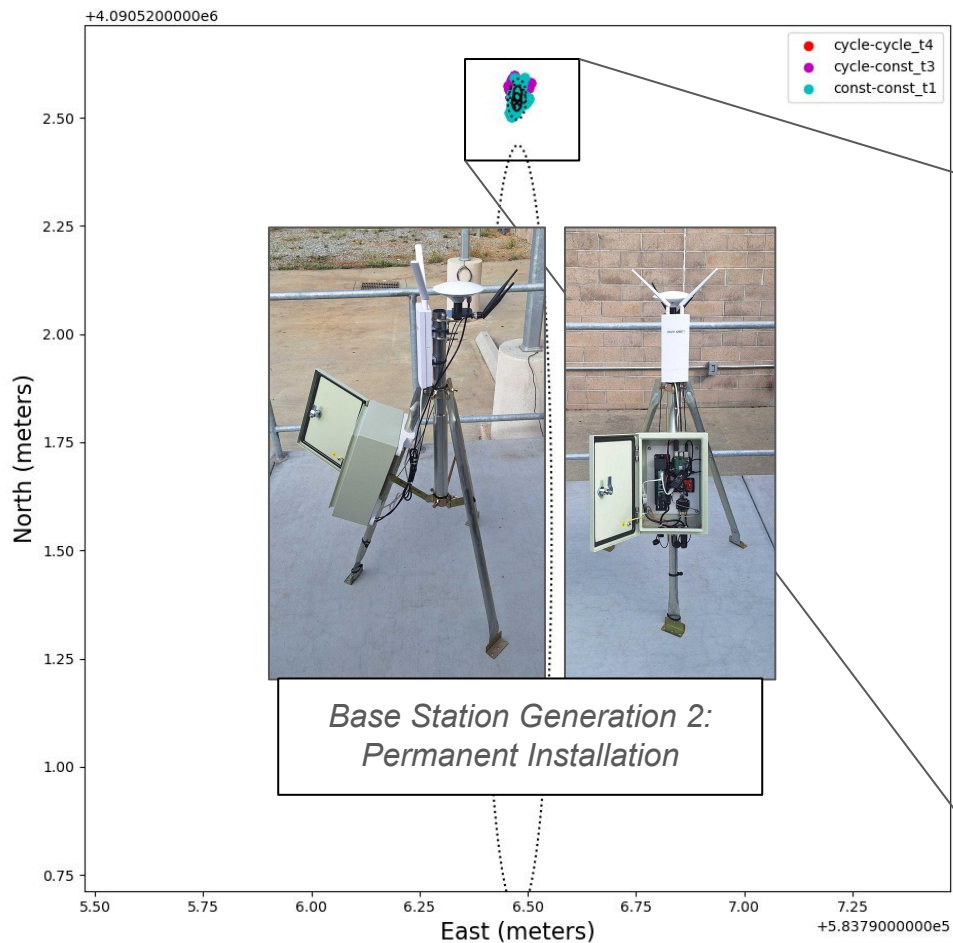
⇒ The solution of a given RTK fix is roughly 100 times more consistent than the solutions of several RTK fixes.

To verify that hypothesis, we repeated treatment 4 (3 sets of 30).

⇒ Once the base station is forced to re-survey, the solutions return to varying on the scale of tens of centimeters, versus the scale of centimeters

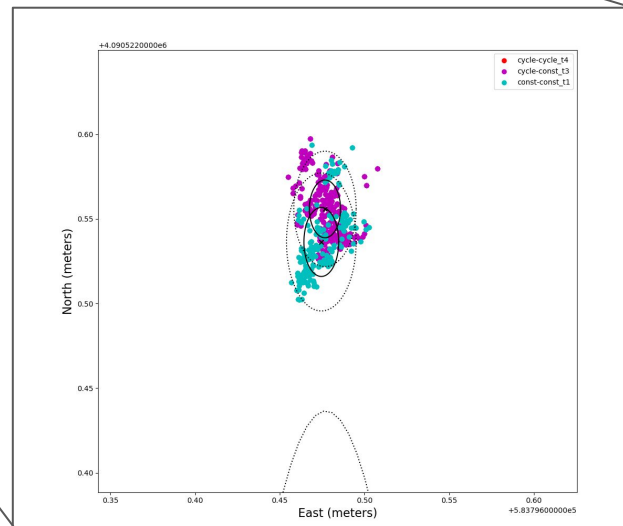


3 cycles @ accuracy threshold 0.10m
not significantly tighter a spread



Why no Treatment 2?

- T3 is faster to perform.
- T2 power cycle is still isolated.
 - ◆ T1 + T3 ⇒ T2 introduces significant variance.



Agricultural Pathology Studies

IDEA:

Use BirdsEye to pursue **much harder** detection and classification problems.

“Realistic and practical.”

“Economically significant.”

Two-Spotted Spider Mite Detection Workflow:

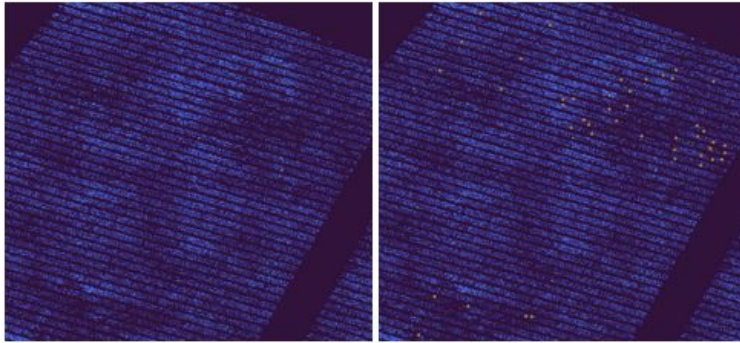


Figure 2. Vignette of a georeferenced VARI orthophoto of a TSSM-affected field. The right panel displays RTK annotations placed at the onset and offset of visually stressed groups of plants, heuristically correlated to more intense TSSR infestations.

Crown Rot Detection:

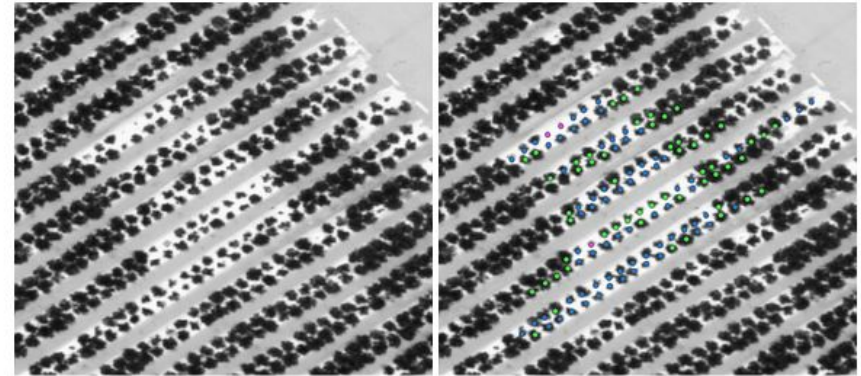


Figure 3. Vignette of a georeferenced NDVI orthophoto showing distinct canopy thinning due to soilborne fungal pathogens. The left panel shows the unannotated orthophoto, while the right panel appends our tightly georeferenced annotations of dead (pink), less vigorous (blue), and vigorous (green) plants in the affected zone. These classes were developed by collaborating field plant pathologists.